Automated Waste Classification using Deep Learning

Project Proposal

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Current situation at KWL

- KWL operates a high-capacity materials recovery facility
- We use mechanical screens, human quality checks, magnets and near-infrared (NIR) sensors for sorting the waste
- Plastic, and especially PET plastic, has a high market price and arrives a lot at KWL
- Despite household pre-sorting, contamination in the “Yellow Bag” is high and our NIR sensors that use light reflection for classification struggle to always correctly identify the polymer type

Project goals and value

- Project goal: integrating a CNN-based vision layer that uses shape, texture and other visual context to augment existing infrastructure
- Subgoal: high precision for classifying transparent plastic
- Value if sucessful:
  - higher prices for high-purity plastic, metal, cardboard and glass
  - reducing the need for human work
  - avoiding fees for not reaching recycling goals (Verpackungsgesetz)
  - real-time statistics for reporting

Project details

- Dataset: over 4700 color images of waste received at the Whyte's Gully Waste and Resource Recovery facility in Wollongong NSW Australia
- Images are sorted into 9 categories: Cardboard, Food Organics, Glass, Metal, Miscellaneous Trash, Paper, Plastic, Textile Trash and Vegatation → the same categories as we have at KWL; I would further label the plastic since it’s important for us to identify transparent plastic
- The images reflect the actual conditions of the waste at a waste recovery facility → dirt, moisture, organic residues; crushed and torn items
- Number of images is sufficient to properly train a CNN

Sources

- https://archive.ics.uci.edu/dataset/908/realwaste
- https://en.wikipedia.org/wiki/Materials_recovery_facility
- https://www.euwid-recycling.de/maerkte/
- https://www.gesetze-im-internet.de/verpackg/__1.html